Knowledge and Language 2025/2026
literature Review

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I. Abstract

This literature review explores the integration of Knowledge Graphs (KGs) and Retrieval-Augmented Generation (RAG) techniques for domain-specific question answering. The challenge we are trying to tackle is building a system that can reliably and accurately answer complex, domain-specific questions about Portuguese cuisine and other recipes using the food.com dataset [1]. We will use RecipeRAG [1], a KG-driven recipe generation framework, as reference for the methodologies to build a food chatbot. Definitions and concepts are supported by the class materials on semantic web framework (RDF, OWL, SPARQL), establishing the link between symbolic and knowledge graph representation.

II. Connections with the Topics Covered by the Course

Large Language Models (LLMs) demonstrate strong generalization but fail in domain-specific reasoning due to three main limitations: (1) lack of contextual understanding, (2) incomplete factual knowledge, and (3) weak recall of precise facts [2]. This motivates the integration of symbolic knowledge (RDF/OWL-based graphs) with neural inference (LLMs and RAG) to ensure both reasoning and factual grounding. RDF can be serialized in formats such as N-TRIPLES, XML, JSON-LD, or Turtle, ensuring interoperability across systems [3]. Vocabularies define terms, relationships and ontologies define classes, hierarchies, and property restrictions. OWL (Web Ontology Language) can be used with RDF to express richer logical constraints, such as domain/range restrictions and class relations [3].
This symbolic foundation aligns with the three “knowledge-aware” dimensions of KG-augmented NLP methods outlined in [2]:

- KG-Augmented Retrieval: The use of structured queries (SPARQL) or textualized triples to retrieve relevant information for LLMs, comparable to RAG pipelines [4].
- KG-Augmented Reasoning and Controlled Generation: Incorporating graph reasoning steps (symbolic inference or logical validation) within the generation loop.
- Knowledge-Aware Training: Using KG-derived data to fine-tune LLMs, injecting factual consistency.
- Knowledge-Aware Validation: Employing SPARQL or rule-based checks to verify the factual correctness of generated answers.

This emphasizes the distinction between symbolic reasoning (FOL-based) and neural reasoning, emphasizing that symbolic models provide interpretability and factual precision, while neural models capture language variability and contextual generation.

III. RDFS/OWL Knowledge Graphs for Representing Knowledge from Datasets

In RDF, knowledge is represented through triples — for example:

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<Bacalhau_a_Brás> <hasIngredient> <Egg>
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Each triple can be enriched through RDFS with class and property hierarchies and constraints. In [5] RecipeKG is constructed as a large-scale recipe knowledge graph combining ingredients, cooking techniques, and regions, and enriching it with semantic annotations. Enabling structured querying and semantic interoperability through URIs and ontologies.
Symbolic KGs are inherently sparse, so in RecipeRAG Knowledge Graph Embeddings (KGEs) are employed to map entities and relations into continuous vector spaces. Foundational models [5] include TransE, RotatE, and QuatE, representing relations as translations, rotations, and quaternion transformations respectively. FOr this project, we will be using Intent extraction via NLP models to map user queries to KG entities and relations. The system uses BERT and spaCy together to understand user queries about recipes. BERT is fine-tuned to identify the user’s intent, such as finding a recipe, getting ingredients, or saving a recipe. This helps the system understand what the user wants, even if the question is phrased in different ways. After the intent is identified, spaCy extracts important entities from the query, such as ingredients, diet types, cuisines, and time limits. These entities are then linked to properties in the knowledge graph. The intent and entities are combined to create a SPARQL query that retrieves relevant recipes

IV. Knowledge Retrieval via NLP

Knowledge retrieval connects natural language input to structured or embedded KG representations. SPARQL is the declarative query language for RDF graphs [3], enabling semantic retrieval using triples and filters. SPARQL queries operate over

RDF graphs, and their construction can be made in 3 different ways:

- The user using the Chatbot need to input the SPARQL query directly or use a form-based interface that generates the query based on user selections.
- LLMs can be fine-tuned or prompted to generate SPARQL queries from natural language questions, using few-shot examples or templates to guide the generation.
- NLP tools like semantic parsers can convert natural language into SPARQL queries by identifying entities, intent, relations, restrictions and query structure.

Natural language interfaces require NLP-based query interpretation, extracting entities and intents (via tools like spaCy or BERT) and mapping them to KG entities (URIs). KGE-based retrieval [5] significantly outperforms text-based retrieval when handling multiple user constraints. As noted: "Performance declined as the number of criteria increased... Nevertheless, the results underscore the effectiveness of KGEs in supporting multi-criteria recipe retrieval, while simultaneously revealing the limitations of purely text-based embeddings." For this project, we will be using embeddings to retrieve the top-k most similar recipes from the textual properties not present in the KG (like recipe name and description).

V. Retrieval-Augmented Generation (RAG)

RAG, first formalized by [4], is a hybrid model that retrieves external documents as the input sequence x to retrieve text documents z and use them as additional context when generating the target sequence y. This approach separates retrieval from generation, improving factuality and reducing hallucinations. While the original RAG used text documents, later adaptations like the mentioned RecipeRAG [5] replace text retrieval with KG-based retrieval.

The foundational work by Lewis et al. [4] introduced RAG models as a combination of parametric and non-parametric memory systems: a pre-trained seq2seq transformer (BART) serves as the parametric memory, while a dense vector index of Wikipedia—accessed through a neural retriever (DPR)—acts as non-parametric memory. This allows the model to retrieve the top-k most relevant documents for each query and marginalize over them when generating the output, effectively blending symbolic retrieval with neural generation.

Two model variants were proposed: *RAG-Sequence*, which conditions generation on a single retrieved document for the entire sequence, and *RAG-Token*, which allows token-level retrieval, providing finer factual grounding. Their experiments demonstrated state-of-the-art performance across knowledge-intensive tasks such as Natural Questions, TriviaQA, and FEVER, outperforming both parametric-only (e.g., T5, BART) and purely extractive retrieval models (e.g., DPR, REALM). Notably, RAG models also produced more specific, diverse, and factually correct generations, showing reduced hallucination and enabling knowledge updates simply by replacing the underlying index.

RecipeRAG [5] builds upon this foundation, substituting text retrieval with KG-based retrieval and applying it to domain-specific recipe generation. The authors state:

"We chose RotatE, the strongest-performing KGE model, to bootstrap the recipe generation process. The system retrieves the top-ranking recipes based on aggregated relevance scores, selects the top 5, and feeds them—together with user criteria—into the generator (DeepSeek-R1-0528)."

This pipeline combines the retrieval mechanism of [4] with structured KG enrichment, turning symbolic knowledge into context for generative models. Given this, we will implement the following layered knowledge representation model: symbolic layer (RDF/OWL) → Query Layer (SPARQL) → neural layer (LLM generation given top 5 most similar recipes).

RAG thus operationalizes the Knowledge-Aware Inference paradigm [2], where external structured knowledge is used to enhance reasoning and factual correctness.

VI. Integrating Concepts for the Proposed Architecture

The proposed architecture is as follows:

KG Construction: Build a Portuguese recipe KG using RDF/OWL. Define *Recipe*, *Ingredient*, *Region*, and *Technique* as OWL classes with property restrictions (e.g., *hasIngredient only Ingredient*).

Entity Recognition: Use spaCy/BERT for entity linking and intent recognition. Map entities to KG URIs through ontology matching and synonym expansion.

SPARQL Verification: Execute SPARQL queries to validate factual attributes (e.g., preparation time, allergens, dietary restrictions).

RAG Generation: Inspired by the RAG-Sequence and RAG-Token formulations from [4], the system retrieves the top-k most relevant recipes (based on KGE similarity or SPARQL-filtered attributes) and uses them as external non-parametric memory for the generative model. This mechanism allows the LLM to incorporate factual culinary knowledge directly into its responses while preserving natural language fluency. Unlike text-only retrieval over Wikipedia, our system will integrate structured RDF triples as retrievable context, unifying symbolic reasoning (SPARQL querying) with neural reasoning (RAG-based language modeling).

Validation: Post-generation KG validation (checking logical consistency via OWL reasoning) ensures that outputs adhere to domain constraints and semantic accuracy.

VII. Conclusion and Future Work

The reviewed literature converges on the same insight: combining symbolic reasoning (RDF, OWL, SPARQL) and neural reasoning (KGE, RAG, LLMs) yields systems that are both factual and context-aware. Empirically, RecipeRAG [5] and RAG [4] demonstrate that KG-driven retrieval enhances generation quality, while the taxonomy of [2] clarifies how such systems reduce hallucination through Knowledge-Aware Inference, Learning, and Validation.

REFERENCES

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